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Geomagnetic Secular Variation During the Brunhes Epoch in the Indian and Atlantic Ocean Regions

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Summary

Brunhes epoch secular variation in the Indian and Atlantic Ocean regions has been examined using available and unpublished palaeomagnetic results from igneous rocks. Specimens from 401 separate units, all of which have unstable components minimized, are involved. More than half of these data are previously unpublished. The data are classified as first order when the mean direction for each body is derived from at least four separate specimens, and second order when the number of specimens per body is not considered. Geomagnetic palaeosecular variation models are examined in terms of the latitude dependence of angular dispersion (S) of virtual geomagnetic poles (VGP's).

Value of S from second order data, while exhibiting wide 95 per cent confidence intervals, increase with latitude. First order data also yield S values increasing with latitude, and are sufficiently well defined to indicate that S may have increased more rapidly with latitude during the Brunhes Epoch than at present. While the simple diode wobble model for the geomagnetic field is clearly excluded by the results, no distinction can be made between other current models, and the possibility of the existence of a region of attenuated Brunhes epoch non-dipole activity in the west-central Indian Ocean cannot be excluded.

Introduction

Spherical harmonic analysis divides the geomagnetic field into two major components: the main geocentric dipole, which is at present inclined at 11½° to the spin axis, and the non-dipole field which at present is best envisaged as being due to 10 to 12 irregularly spaced small dipoles in, and normal to, the surface of the outer core. The non-dipole field constitutes, depending on geographic position, from 0 to 30 per cent of the geomagnetic field at the Earth's surface. Observatory data show that the main dipole has been stationary for at least the past 150 years, although it is decreasing in intensity at about 6 per cent century, whereas the non-dipole field moves and changes intensity rapidly. Those latter movements cause the geomagnetic secular variation. The best estimates of non-dipole movement for the present century is westward drift at about 0·2° of longitude per year. One of the major known characteristics of the non-dipole field is the regional minima over most of the central Pacific (Cox 1962). Here the effect of the decreasing main dipole is evident, but the absence

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of the non-dipole field is expressed in virtually unchanging directions of the geomagnetic field, or very low secular variation.

Knowledge of geomagnetic field behaviour prior to the initiation of observatory measurements must be obtained indirectly, by the palaeomagnetic method. Similarly, palaeosecular variation can be examined, if palaeomagnetically testable models are available. Models of the geomagnetic field expressed as relations between secular variation and geographic latitude have been formulated. Latitude dependence of any geomagnetic directional behaviour results largely because of the two-fold increase of geomagnetic field intensity between the equator and axis of a centred dipole. Models can be expressed as functions of the geomagnetic field directions (declinations and inclinations) or the corresponding virtual geomagnetic poles (or co-ordinates of the surface expression of the geocentric dipole giving rise to the observed directions). The latter presentation convention has several advantages over the former, when geomagnetic field models are analysed (Cox 1969). The secular variation is best expressed as S in

$$S^2 = \frac{1}{N-1} \sum_{i=1}^{N} (\delta i)^2 \tag{1}$$

where N = the number of observations, and δi = the angular distance between the virtual geomagnetic pole (VGP) resulting from a given D and I at the site location, and true mean VGP. The true mean direction for the Brunhes Epoch is probably at, or very close to, the present geographic pole. The time span of the observations should be sufficient to sample both dipole and non-dipole movements: this period would appear to be of the order of 10^4 years. The models of geomagnetic field behaviour which have so far been formulated for palaeomagnetic testing are:

Model A. Vectors of varying orientation but constant amplitude (to simulate non-dipole activity) are added to a fixed axial dipole (Irving & Ward 1964). The variation of S with latitude is given in Fig. 1.

Model B. Creer, Irving & Nairn (1959) proposed a model due to wobble of the main dipole only, without any non-dipole contribution. If resulting field directions (D and I) are analysed directly, the dispersion of directions vary latitudinally: but if the corresponding VGP are analysed, the effect of dipole wobble is the same at all latitudes (Fig. 1).

Model C. This model combines dipole wobble and a randomly varying non-dipole component. In Fig. 1 this is shown, using an 11½° maximum dipole wobble (S_D) which Doell (1969) has shown to be the most appropriate to the Brunhes epoch (t = 0 to 0·7 My), and the present non-dipole behaviour (S_N). The value of S_N at each latitude is obtained by combining the VGP for the non-dipole field at each 10° longitude interval and averaging between hemispheres. The total secular variation (S_T) is then obtained from

$$S_T = (S_D{}^2 + S_N{}^2)^{\frac{1}{2}} \tag{2}$$

Doell & Cox (1971, 1972) have applied Brunhes epoch palaeomagnetic data from the Pacific region to these models, with the results shown in Fig. 2. The low value of S for the Hawaiian Islands, shows that present low values of secular variation in the central Pacific have persisted since at least t = 0·7 My. The favoured cause for this phenomenon is the existence of a major mantle inhomogeneity coupled to the Earth's core so that non-dipole fields are suppressed, or shielded out at the Earth's surface.

It is clearly desirable to extend these observations to other regions, to assist in defining the limits of the proposed mantle inhomogeneity, in addition to further

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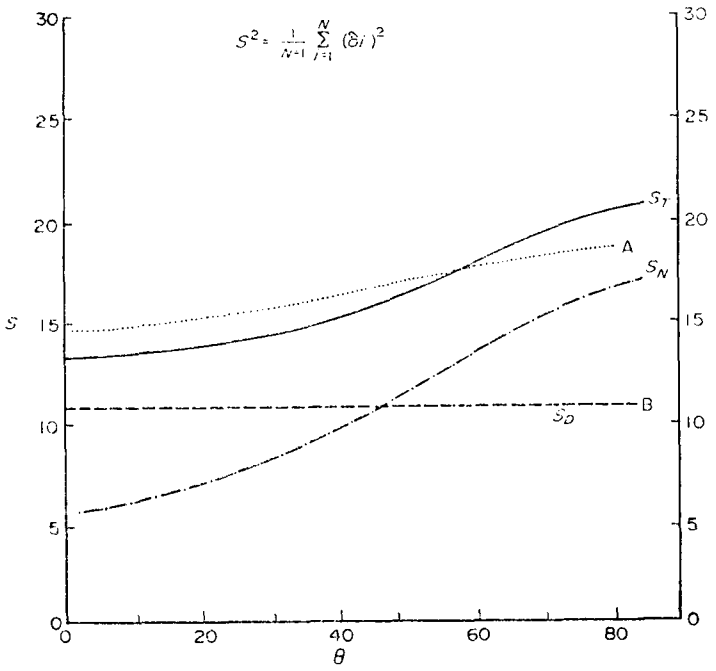


FIG. 1. Relation between geographic latitude (θ) and angular dispersion (S) of virtual geomagnetic poles (VGP) for four different geomagnetic field models. Equation relating S to number of sites (N) and angular difference been VGP and north geographic pole (δi) given. S_D and B = field due to dipole wobble of 11°. S_N = field due to 1945 IGRF non-dipole field, averaged at 10° longitudinal intervals, and averaged between hemispheres. A = model due to varying orientation but fixed amplitude vector added to constant axial dipole field. S_T = total field behaviour for 1945 IGRF Field. See text for sources of models.

Note S_T = model C (see text).

defining past relationships of S and latitude. According to the 1945 International Geomagnetic Reference Field, the non-dipole field intensity is low in the west-central Indian Ocean and other regions. The existence of non-dipole attenuation in addition to the central Pacific feature is therefore worthy of consideration.

In this communication we extend Brunhes epoch secular variation studies to the Indian and Atlantic Ocean regions.

Geological control

As Cox (1969) has stressed, the substantial difficulties usually involved in assigning geological ages to a series of igneous rocks are greatly minimized for Brunhes epoch (t = 0 to 0·69 My) materials. Outcrops of this age can easily be distinguished from the older Matuyuma epoch materials using polarity differences. Geomorphological criteria, and very high latitude VGP assist in this identification. Nevertheless, potassium–argon age determinations can increase the confidence of such age assignments.

We are presenting a compilation of Brunhes epoch paleomagnetic results from igneous rocks of the Indian and Atlantic Ocean regions. Many of the data have already been presented. We describe below the geological control only for those sampling sites previously unreported. The geographic co-ordinates of each body are included in Table 1. The geological control for the remaining results are given in the respective references.

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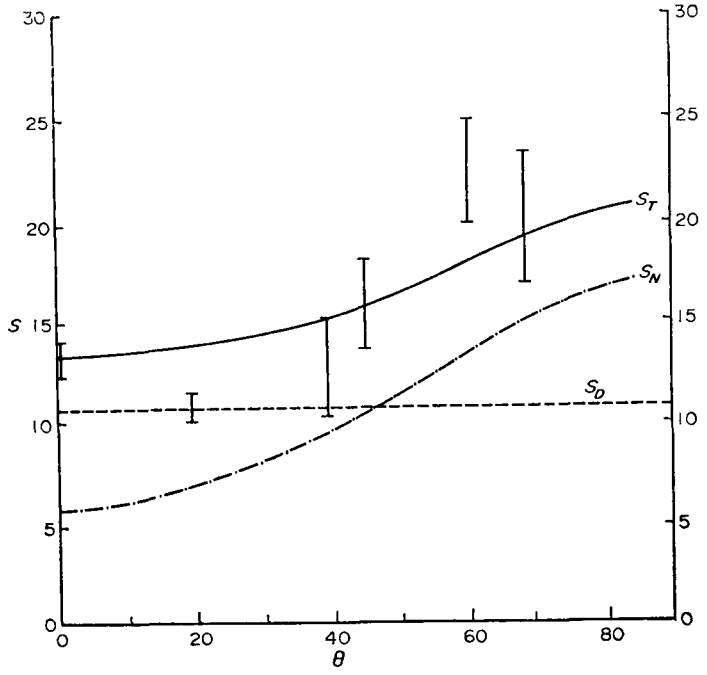


FIG. 2. Values of S in degrees for the Pacific Region (Doell & Cox 1971), as a function of latitude (θ). Results are given in form of 95 per cent confidence limits of S. See Fig. 1 and text for explanation of S_D, S_N, and S_T. Geographic sources of data, in order of increasing latitude are Galapagos Islands; Hawaiian Islands; New Zealand: Western USA; Alaska; Antarctica. See Doell & Cox (1972) for fine details. Latitude in degrees. Hemispheres not distinguished.

The Comore Islands

Four principal islands comprise the Comores, which are located at the northern end of the Mozambique Channel (Fig. 3). Esson *et al.* (1970) have recently reviewed the geology of the islands. Brunhes epoch lavas have been sampled on Grande Comore and Anjouan.

Grande Comore is the largest of the Comore Islands, and is geomorphologically young, being without any deeply incised valleys. It is dominated by the volcanic centres of Massif del a Grille (rising to 1087 metres above sea level) and Le Kartala (2361 metres), in the north and south respectively. Five of the sampled lava flows are from fissures of the northern centre. The possible tectonic significance of sandstone inclusions in these lavas have been discussed by Strong & Flowers (1969). Le Kartala has experienced Historical eruptions from both the summit and flank fissures. Nine lavas, sampled on the west and central coasts, are ankaramites or feldsparphyric flows from fissure eruptions of La Kartala. All lavas are from the ' Phase Superieur ' stratigraphic subdivision of de Saint Ours (1960), which forms most of the prominent terrace of 8 m height fringing the coast.

Twenty-three lava flows were sampled on Anjouan. The topography is more dissected than on Grand Comore, consistent with an older age for the island. Lavas 2, 6, 7, 17, 18 and 21 are from the melanocratic-ankaramitic volcanic series of de Saint Ours (1960). These underly the other lavas sampled which are all from the Basaltic Volcanic Series. The older parts of the island, which have not been sampled, have been deeply eroded, providing channels for the younger sampled lavas, which are highly alkalic and feldspathoidal, including basanites and trachybasalts.

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Table 1

Brunhes epoch palaeomagnetic data (all previously unpublished)

(i) Indian Ocean

(a) Comores Islands (Anjouan)

Unit	ϕ° E	θ° S	N	D	I	R	θ'	ϕ'	$\alpha(95)$
AJ01	44·45	12·17	6	358·5	−30·1	5·990	85·8	244·4	2·95
AJ02	44·47	12·18	6	16·2	−30·4	5·947	73·8	151·1	6·96
AJ03	44·53	12·27	7	356·2	−17·7	6·987	85·1	354·0	2·78
AJ04	44·50	12·27	6	357·8	02·0	5·762	76·6	34·9	14·97
AJ05	44·52	12·52	2	357·7	−15·1	1·994	84·9	17·6	19·35
AJ06	44·40	12·17	6	4·6	−06·5	5·995	80·0	71·8	2·16
AJ07	44·40	12·17	6	18·4	−13·2	5·976	71·0	119·2	4·61
AJ08	44·38	12·20	4	5·3	−28·5	3·988	84·0	165·1	5·92
AJ09	44·38	12·20	8	11·5	−20·5	7·966	78·6	127·4	3·80
AJ10	44·38	12·20	7	17·3	−23·8	6·961	73·1	137·1	4·97
AJ11	44·38	12·20	5	7·0	−17·3	4·997	82·3	109·1	2·21
AJ12	44·38	12·20	5	11·7	−19·2	4·965	78·3	124·1	7·15
AJ13	44·28	12·20	4	348·0	−34·6	3·917	76·6	282·3	15·47
AJ14	44·28	12·20	4	0·7	−12·3	3·966	84·0	51·1	9·87
AJ15	44·28	12·20	5	350·4	−41·9	4·971	75·0	260·3	6·53
AJ16	44·28	12·20	8	347·2	−46·0	7·966	70·7	260·8	3·88
AJ17	44·43	12·30	8	10·9	−14·5	7·950	78·2	111·2	4·69
AJ18	44·43	12·30	6	16·8	−11·2	5·925	72·2	114·3	8·27
AJ19	44·40	12·17	3	2·3	−15·7	2·999	85·2	72·9	3·21
AJ20	44·40	12·17	4	34·7	−19·7	3·484	55·9	134·5	41·74
AJ21	44·38	12·18	9	351·1	−35·3	8·107	78·8	272·8	18·20
AJ22	44·43	12·13	8	359·0	−21·4	7·976	88·5	3·0	3·25
AJ25	44·35	12·25	4	0·3	−19·0	3·965	87·6	51·3	10·02
AJ26	44·34	12·25	7	335·2	−24·0	6·879	65·8	310·9	8·67
AJ27	44·34	12·25	4	6·8	−19·3	3·995	83·0	116·3	3·89

(b) Comores Islands (Grande Comore)

Unit	ϕ° E	θ° S	N	D	I	R	θ'	ϕ'	$\alpha(95)$
GC01	43·25	11·63	6	8·6	−17·2	5·899	81·1	115·2	9·63
GC02	43·25	11·60	6	344·2	−36·0	5·947	72·7	282·8	6·95
GC03	43·26	11·50	6	0·3	−28·0	5·990	86·8	218·1	3·05
GC04	43·25	11·63	6	1·1	−10·8	5·981	83·7	53·3	4·13
GC05	43·26	11·50	7	354·6	−19·0	6·981	84·4	332·8	3·43
GC06	43·26	11·50	6	352·7	−19·5	5·856	82·6	325·6	11·52
GC07	43·26	11·65	6	347·8	−07·9	5·995	75·7	345·0	2·12
GC08	43·26	11·65	5	353·9	−15·0	4·996	82·7	346·8	2·48
GC09	43·26	11·65	5	357·1	−17·6	4·982	86·1	356·3	5·20
GC10	43·26	11·65	5	5·2	−19·4	4·990	84·6	115·3	3·91
GC11	43·26	11·65	6	355·8	−21·8	5·966	85·9	318·3	5·56
GC12	43·28	11·40	7	2·0	−28·3	6·995	86·1	193·5	1·74
GC13	43·22	11·76	8	18·0	−37·6	7·206	70·4	164·1	19·75
GC14	43·25	11·70	5	18·5	−21·3	4·151	71·9	133·0	39·47

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Table 1 (*continued*)

(c) Amsterdam Island

Unit	ϕ° E	θ° S	N	D	I	R	θ'	ϕ'	$\alpha(95)$
NA01	77·54	37·78	6	3·7	−58·6	5·996	86·7	196·5	2·29
NA02	77·54	37·77	8	70·4	−48·7	7·560	32·2	182·1	20·36
NA03	77·54	37·77	3	18·7	−66·6	2·990	72·4	213·6	1·66
NA04	77·54	37·80	8	25·3	−66·3	7·994	68·7	206·6	2·38
NA05	77·52	37·78	6	13·7	−32·3	5·971	66·4	112·0	6·18
NA06	77·52	37·78	6	14·0	−33·3	5·934	66·9	113·0	9·28
NA07	77·52	37·78	6	344·4	−48·9	5·974	74·8	14·7	5·87
NA08	77·52	37·78	6	4·1	−54·8	5·976	85·9	131·9	5·57
NA09	77·52	37·78	6	16·9	−57·6	5·971	76·7	174·7	6·14
NA10	77·52	37·78	6	18·8	−67·5	5·994	71·7	216·8	2·88
NA11	77·51	37·87	6	343·5	−30·8	4·928	64·3	38·7	38·11
NA12	77·50	37·88	8	345·1	−60·9	7·943	77·9	323·0	7·30

(d) Crozet Islands (Possession)

Unit	ϕ° E	θ° S	N	D	I	R	θ'	ϕ'	$\alpha(95)$
CP01	51·85	46·43	6	21·7	−62·4	5·999	74·5	139·3	1·12
CP02	51·78	46·44	4	38·1	−67·0	3·983	64·6	163·0	6·99
CP03	51·78	46·44	5	32·7	−38·7	4·849	53·8	109·0	15·12
CP06	51·86	46·42	6	51·7	11·1	5·194	20·7	108·3	29·22
CP09	51·79	46·45	4	354·8	−62·5	3·992	85·5	355·7	4·78
CP10	51·79	46·45	6	2·9	−71·6	5·991	80·0	222·4	2·87
CP11	51·79	46·45	4	0·7	−71·8	3·993	79·8	229·5	4·34
CP12	51·79	46·45	4	12·0	−59·9	3·994	79·6	112·6	4·13
CP13	51·79	46·45	6	11·4	−55·7	5·981	76·7	95·4	4·16
CP20	51·78	46·47	4	354·2	−70·4	3·974	81·2	254·1	8·65
CP21	51·78	46·47	4	5·0	−66·3	3·979	86·0	176·9	7·64
CP22	51·78	46·47	4	4·3	−63·0	3·997	86·4	109·2	2·96
CP23	51·78	46·47	4	5·5	−66·4	3·997	85·6	176·2	2·88
CP24	51·78	46·47	4	345·3	−67·0	3·991	79·7	298·4	5·16
CP25	51·78	46·47	4	351·8	−64·2	3·998	84·3	324·1	2·44
CP26	51·78	46·47	4	10·0	−64·5	3·997	83·1	144·1	2·92
CP27	51·78	46·47	5	359·8	−61·3	4·984	85·9	49·6	4·89
CP28	51·78	46·47	4	348·0	−55·7	3·995	76·4	6·3	3·81
CP29	51·78	46·47	3	334·5	−51·5	2·969	65·8	349·0	15·36
CP30	51·78	46·47	6	356·3	−56·2	5·968	79·9	34·6	5·40
CP31	51·78	46·47	4	32·9	−71·7	3·995	67·5	180·2	3·59
CP32	51·78	46·47	4	358·0	−75·6	3·983	73·6	234·9	6·91
CPA32	51·78	46·47	5	16·0	−77·1	4·993	69·2	212·8	3·16
CP33	51·78	46·47	4	349·4	−68·8	3·982	81·1	278·2	7·09
CP34	51·78	46·47	4	15·2	−51·8	3·982	71·7	96·7	7·15
CP35	51·78	46·47	6	333·8	−48·0	5·945	61·1	353·0	7·04
CP36	51·78	46·47	6	329·5	−49·2	5·960	61·3	345·7	6·03
CP37	51·78	46·47	4	356·7	−38·5	3·790	65·0	44·4	25·20
CP38	51·78	46·47	6	343·0	−69·6	5·984	77·7	289·4	3·76
CP39	51·78	46·47	6	340·3	−58·4	5·981	73·8	341·8	4·12
CP41	51·82	46·37	4	356·8	−66·2	3·993	87·0	276·5	4·38
CP42	51·82	46·37	6	349·2	−70·8	5·932	79·0	265·9	7·89
CP43	51·82	46·37	6	356·6	−71·4	5·926	80·2	242·9	8·22
CP44	51·82	46·37	6	340·9	−54·5	5·989	71·6	353·5	3·12
CP45	51·82	46·36	6	10·3	−63·6	5·989	82·7	135·2	3·20
CP46	51·82	46·36	2	13·7	−57·9	1·998	77·2	108·3	10·18
CP47	51·82	46·36	5	52·7	−65·6	4·955	54·8	163·6	8·17
CP48	51·83	46·36	6	21·4	−50·8	5·546	67·7	106·9	21·11

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(e) Crozet Islands (East Island)

Unit	ϕ° E	θ° S	N	D	I	R	θ'	ϕ'	$\alpha(95)$
CE01	52·28	46·41	6	343·6	−76·0	5·993	70·7	254·6	2·58
CE02	52·28	46·41	5	40·8	−74·5	4·451	62·5	188·8	30·41
CE03	52·18	46·39	4	17·5	−48·5	3·997	68·2	97·0	2·91
CE04	52·18	46·39	4	359·5	−49·3	3·926	73·7	50·7	14·65
CE05	52·18	46·39	4	9·9	−62·0	3·991	82·3	120·8	5·08
CE06	52·18	46·39	4	0·6	−55·0	3·994	79·0	54·8	4·08
CE07	52·18	46·39	4	5·6	−59·0	3·955	82·1	85·4	11·35
CE08	52·18	46·39	6	18·7	−62·3	5·913	76·5	136·6	8·89
CE09	52·18	46·39	6	24·2	−68·9	5·989	73·3	171·5	3·18
CE10	52·18	46·39	7	348·8	−68·7	6·982	80·8	280·8	3·29
CE11	52·25	46·40	4	10·8	−61·8	3·988	81·6	121·5	5·73
CE12	52·24	46·40	5	8·0	−69·2	4·925	81·9	195·6	10·58
CE13	52·24	46·40	4	11·7	−60·0	3·983	79·9	112·8	7·00
CE14	52·24	46·40	4	360·0	−63·5	3·992	88·6	52·2	4·86
CE15	52·24	46·40	4	359·1	−61·1	3·989	85·6	43·4	5·47
CE16	52·24	46·40	4	14·1	−64·8	3·998	80·3	148·7	2·44
CE17	52·24	46·40	4	7·4	−63·0	3·985	84·4	123·4	6·44
CE18	52·24	46·40	4	28·7	−59·3	3·861	68·3	135·5	20·22
CE19	52·24	46·40	4	14·2	−54·7	3·970	74·5	100·7	9·30
CE20	52·24	46·40	5	27·7	−52·9	4·935	65·3	120·3	9·86
CE21	52·24	46·40	4	23·5	−52·1	3·994	67·4	113·0	4·15
CE22	52·24	46·40	4	36·1	−44·9	3·983	55·2	119·8	6·87
CE23	52·25	46·40	4	23·4	−49·2	3·986	65·5	108·3	6·34
CE24	52·24	46·48	4	19·0	−52·6	3·986	70·3	106·3	6·29
CE25	52·24	46·48	4	5·1	−72·1	3·973	78·9	217·7	8·77
CE26	52·24	46·48	4	338·3	−59·8	3·752	73·3	334·8	27·57
CE27	52·24	46·48	4	47·1	−61·8	3·976	56·9	153·6	8·27
CE28	52·24	46·48	4	43·9	−69·7	3·997	61·4	172·7	2·83
CE29	52·24	46·48	4	5·8	−64·2	3·999	86·0	136·7	1·35
CE30	52·24	46·48	4	30·6	−51·5	3·995	62·6	121·6	3·91
CE31	52·24	46·48	4	33·0	−50·0	3·998	60·1	122·2	2·36
CE32	52·24	46·48	4	1·2	−41·8	3·925	67·6	55·1	14·70
CE33	52·24	46·48	4	3·6	−53·7	3·986	77·4	66·0	6·18
CE34	52·24	46·48	4	350·8	−59·3	3·999	80·7	2·7	1·52
CE35	52·24	46·48	4	357·3	−58·2	3·995	82·1	36·7	3·80
CE36	52·24	46·48	4	5·4	−53·5	3·998	76·9	72·3	2·53
CE37	52·24	46·48	4	352·3	−56·9	3·990	79·3	17·1	5·25
CE38	52·24	46·48	4	346·2	−58·2	3·993	77·6	353·1	4·36
CE39	52·24	46·48	4	331·0	−32·1	3·941	52·3	3·0	13·02
CE40	52·24	46·48	4	314·3	−70·7	3·993	60·5	288·6	4·47
CE41	52·24	46·48	4	318·0	−74·1	3·998	62·1	277·2	2·22
CE42	52·24	46·48	4	295·9	−76·9	3·992	51·7	269·9	4·62
CE43	52·24	46·48	4	302·8	−77·1	3·968	54·6	269·3	9·61

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Table 1 (*continued*)

(ii) Atlantic Ocean

(a) Canary Islands (Lanzarote)

Unit	ϕ° W	θ° N	N	D	I	R	θ'	ϕ'	$\alpha(95)$
LZ13	13·47	29·13	2	356·2	51·9		85·3	303·8	—
LZ14	13·47	29·13	2	0·4	40·0	—	83·7	163·2	—
LZ15	13·47	29·13	2	18·4	46·1	—	73·7	77·9	—
LZ16	13·47	29·13	2	348·1	26·4	—	71·3	205·0	—
LZ17	13·47	29·13	2	352·4	34·9	—	77·9	203·2	—
LZ19	13·47	29·13	2	9·3	34·5	—	76·8	124·5	—
LZ20	13·47	29·13	2	21·0	38·0	—	69·5	94·0	—
LZ21	13·47	29·13	2	22·7	30·5	—	65·6	102·9	—
LZ22	13·47	29·13	2	25·6	32·4	—	63·9	97·0	—
LZ23	13·47	29·13	2	27·8	30·4	—	61·5	96·9	—
LZ24	13·47	29·13	2	24·2	36·3	—	66·3	93·2	—
LZ25	13·47	29·13	2	28·6	32·7	—	61·5	93·5	—
LZ26	13·47	29·13	2	26·2	31·5	—	63·1	97·4	—
LZ27	13·47	29·13	2	28·6	34·5	—	62·1	91·3	—
LZ28	13·47	29·13	2	22·7	37·0	—	67·8	93·7	—
LZ29	13·47	29·13	2	26·7	34·6	—	63·7	93·0	—
LZ30	13·47	29·13	2	31·7	35·7	—	59·8	87·3	—
LZ30	13·47	29·13	2	31·7	35·7	—	59·8	87·3	—
LZ31	13·47	29·13	2	24·6	30·5	—	64·1	100·4	—
LZ107	13·80	28·93	5	22·1	33·4	4·961	67·1	99·9	7·57
LZ111	13·50	28·97	2	14·8	61·2		72·2	24·7	
LZ112	13·63	29·05	7	8·2	45·9	6·988	82·6	88·6	2·86

(b) Canary Islands (Gran Canaria)

Unit	ϕ° W	θ° N	N	D	I	R	θ'	ϕ'	$\alpha(95)$
GC28	15·55	28·00	2	3·4	32·0	—	78·9	147·2	—
GC29	15·54	28·01	2	12·9	−06·9	—	56·2	140·7	—
GC30	15·54	28·01	2	44·9	16·1	—	43·3	90·5	—
GC31	15·54	28·01	2	59·6	05·2	—	27·9	87·2	—
GC32	15·54	28·01	2	354·9	40·3	—	83·2	207·9	—

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(c) Canary Islands (Tenerife)

Unit	ϕ° W	θ° N	N	D	I	R	θ'	ϕ'	$\alpha(95)$
TF05	16·62	28·17	2	11·9	30·2	—	73·6	119·2	—
TF06	16·62	28·16	2	344·2	06·1	—	60·5	197·3	—
TF07	16·62	28·16	2	351·7	36·5	—	78·9	208·6	—
TF08	16·62	28·16	2	12·3	30·0	—	73·3	118·5	—
TF09	16·62	28·26	2	4·0	23·6	—	73·5	149·9	—
TF65	16·35	28·53	6	354·4	35·5	5·964	79·9	195·2	5·73
TF66	16·35	28·53	3	345·8	44·7	2·987	77·2	247·8	9·79
TF67	16·40	28·43	6	10·8	59·0	5·950	75·6	19·1	6·74
TF68	16·49	28·35	4	0·3	40·6	3·990	84·8	160·8	5·40
TF69	16·53	28·30	4	339·7	50·8	3·993	72·2	268·8	4·38
TF73	16·60	28·40	6	21·5	−45·2	5·372	31·1	141·3	25·29
TF75	16·65	28·39	4	352·6	26·1	3·987	73·8	190·5	5·96
TF77	16·72	28·35	4	348·6	38·6	3·993	77·7	223·6	4·35
TF88	16·63	28·34	5	343·1	37·3	4·991	72·9	231·4	3·72
TF104	16·60	28·20	4	19·5	21·0	3·773	64·7	113·7	26·23
TF112	16·65	28·19	5	339·0	31·4	4·861	67·6	227·7	14·49
TF114	16·63	28·17	2	9·0	23·1	—	71·6	134·8	—
TF115	16·63	28·16	3	6·1	28·6	2·985	75·7	139·3	10·72
TF116	16·55	28·10	4	356·6	−14·3	3·876	54·2	169·6	19·06
TF118	16·52	28·13	3	16·3	33·1	2·999	71·8	105·0	2·52
TF119	16·49	28·17	6	6·9	47·7	5·969	83·9	68·5	5·30
TF121	16·30	28·43	5	4·0	51·9	4·959	84·6	22·7	7·74

(d) Canary Islands (La Palma)

Unit	ϕ° W	θ° N	N	D	I	R	θ'	ϕ'	$\alpha(95)$
LP01	17·90	28·65	5	21·5	47·1	4·983	71·1	68·4	5·01
LP05	17·82	28·70	6	332·7	55·3	3·974	65·9	276·6	54·43
LP06	17·82	28·70	4	359·1	42·8	3·999	86·1	174·3	1·56
LP07	17·80	28·65	3	340·9	52·3	2·878	73·1	271·5	31·43
LP13	17·85	28·59	5	40·1	47·5	4·987	55·0	62·5	4·33
LP15	17·80	28·47	6	0·5	29·9	5·902	77·3	160·1	9·47
LP18	17·79	28·51	7	0·8	38·4	6·982	82·9	156·3	3·34
LP22	17·73	28·63	4	2·8	39·8	3·988	83·4	139·1	5·91
LP23	17·73	28·82	6	0·5	32·0	5·981	78·6	159·9	4·14
LP24	17·73	28·80	4	7·4	48·4	3·981	83·5	64·5	7·33
LP27	17·72	28·78	6	3·1	50·4	5·823	86·4	29·2	12·84
LP28	17·71	28·75	6	15·1	38·8	5·766	74·8	95·4	14·82
LP29	17·70	28·72	4	12·7	23·5	3·989	69·7	123·9	5·49
LP30	17·70	28·69	5	352·1	10·1	4·348	65·2	181·4	33·62
LP31	17·71	28·69	6	344·8	33·2	5·983	72·5	218·4	3·88
LP32	17·73	28·68	6	342·9	45·6	5·924	74·8	250·1	8·34

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Table 1 (*continued*)

(e) Canary Islands (Hierro)

Unit	$\phi°$ W	$\theta°$ N	N	D	I	R	θ'	ϕ'	$\alpha(95)$
H01	17·99	27·72	2	16·1	52·1	—	75·2	48·2	—
H02	17·99	27·75	2	35·9	16·5	—	51·0	94·7	—
H03	17·99	27·75	2	31·5	13·8	—	53·7	100·8	—
H05	17·99	27·75	2	335·9	67·4	—	61·0	309·4	—
H06	18·04	27·72	2	353·9	67·2	—	67·3	331·8	—
H07	18·04	27·72	2	356·2	10·1	—	67·1	171·8	—
H08	18·03	27·72	2	48·8	57·0	—	48·2	45·5	—
H09	18·03	27·72	2	12·6	52·6	—	77·8	42·1	—
H10	18·03	27·71	2	20·4	57·1	—	70·2	36·5	—
H11	18·03	27·71	2	353·8	64·9	—	70·2	329·4	—
H12	18·03	27·71	2	349·5	32·5	—	76·1	208·2	—
H13	18·03	27·71	2	353·0	50·1	—	83·1	281·2	—
H14	17·93	27·77	4	355·8	46·9	3·975	86·3	259·4	8·41
H15	17·93	27·78	2	340·8	44·8	—	72·9	252·2	—
H16	17·93	27·78	2	0·6	43·4	—	87·5	149·2	—
H17	17·93	27·78	2	334·3	49·1	—	67·4	264·0	—
H18	17·93	27·78	2	346·2	39·4	—	76·4	231·8	—
H20	17·92	27·79	2	18·9	41·0	—	72·5	81·6	—
H21	17·92	27·79	4	346·5	51·9	3·899	77·4	277·7	17·18
H22	17·92	27·79	2	38·0	44·1	—	56·2	66·2	—
H23	17·92	27·79	2	10·3	49·0	—	80·7	55·8	—
H24	17·92	27·78	2	10·0	41·6	—	80·2	92·4	—
H25	17·92	27·78	2	21·7	57·3	—	69·2	37·1	—
H33	17·91	27·79	2	17·4	63·9	—	67·4	15·0	—
H34	17·90	27·80	2	41·5	57·7	—	53·9	44·0	—
H35	17·93	27·77	8	326·1	53·1	7·843	60·4	272·1	8·40
H36	17·92	27·79	6	1·1	53·3	5·990	83·8	350·5	2·96
H37	17·92	27·76	4	7·0	56·7	3·992	78·8	11·8	4·83
H38	17·99	27·73	4	26·7	54·0	3·964	66·2	48·6	10·12
H39	17·99	27·73	6	8·7	51·4	5·940	81·3	39·8	7·38
H40	17·99	27·73	4	10·2	37·0	3·982	78·3	107·0	7·18
H41	17·99	27·73	4	9·1	46·7	3·999	81·9	68·1	1·33
H42	17·99	27·73	3	7·7	38·4	2·949	80·7	11·3	20·02
H44	18·00	27·75	6	22·1	06·2	5·951	57·6	117·5	6·64

Unit = Original sampled body number; $\phi°$ = longitude of site (°E or °W); $\theta°$ = latitude of site (°S or °N); N = number of separate oriented samples; D and I = mean declination and inclination of remanent magnetism. D is in degrees east of north; and I is in degrees with respect to the horizontal (no sign = positive or below horizontal; negative = above horizontal); R = resultant vector, applying unit vector per specimen; θ' and ϕ' = latitude and longitude of virtual geomagnetic pole resulting from D and I at given sample sites; $\alpha(95)$ = semi vertical angle of 95 per cent confidence cone (Fisher 1953).

Note: D, I, R, θ', ϕ', and $\alpha(95)$, all correspond to minimum dispersion computation (Irving *et al.* 1961): see text.

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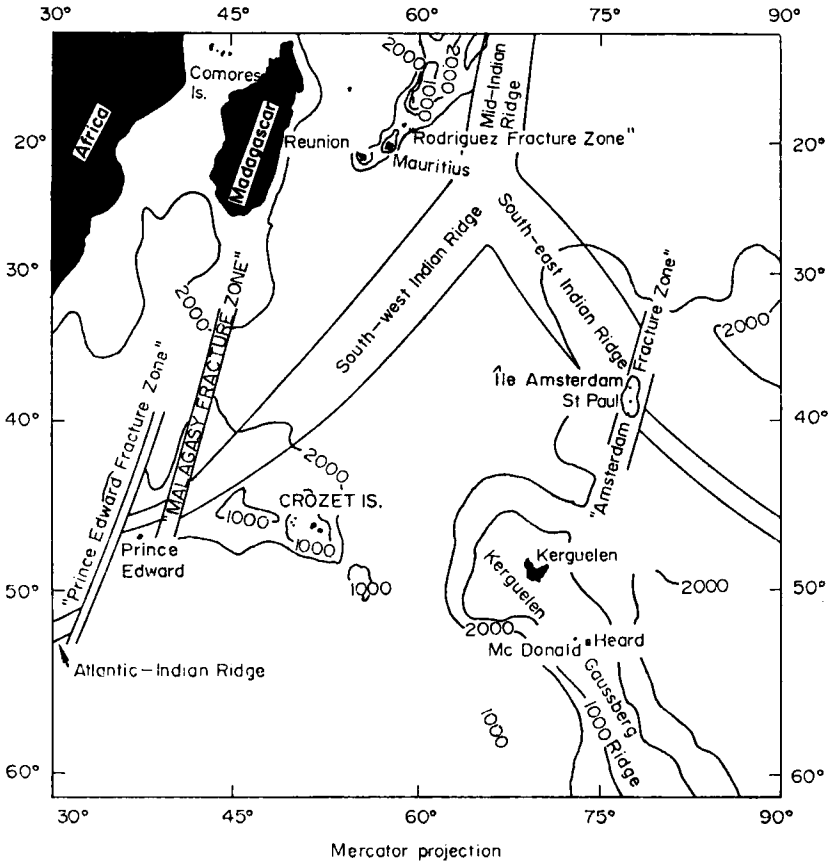


FIG. 3. Map showing locations of south Indian Ocean sampling sites involving previously unpublished results (Comores, Amsterdam, Crozet).

At this time, no radiometric age control is available. We use polarity, VGP latitude, and geomorphological arguments to place the sampled bodies in the Brunhes epoch.

Amsterdam Island

This small island (Fig. 3) lies on a transform fault, as indicated by offsets in the marine magnetic anomaly pattern associated with the southeast Indian Ridge (Dr R. Schlich, private communication). It is a classically formed volcanic cone of very young appearance. Gunn *et al.* (1971) have described the geomorphology of the island, and the locations, petrography, and geochemistry of the 12 sampled lava flows. No radiometric ages are available, but no doubt exists about the young age of the lavas.

Crozet Islands

East Island is a deeply dissected volcanic shield of flat-lying basalts of alkalic affinities, overlying a deeply weathered basement dominated by fine irregular basaltic dykes and sills. Most of the 43 sampled lava flows occur in a single sequence. Gunn *et al.* (1970) have described the geomorphology of the island, and the petrography and geochemistry of the sampled lavas. Locations and cross-sectional diagrams of the three sampled lava sequences are included in Gunn *et al.* (1970).

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Possession island is the largest of the Crozet group. Gunn *et al.* (1972) have compared the petrography and geochemistry of Possession and East Island. While monzonite and syenite have been identified in outcrops on Possession, the island is otherwise very similar geochemically and geomorphologically to East Island, with a series of horizontal to near horizontal lava flows constituting most of the island's surface. No old basement or sub-basalt unconformity is seen on Possession.

Age control on the Crozet Islands is by palaeomagnetic and potassium–argon methods. As will be shown below, all lavas sampled on East Island are of normal polarity. A reversed to normal polarity boundary was detected in the lower parts of three of the sequences sampled on Possession (Morne Rouge, Port Alfred, and Crique de Noel) and this was identified as the Brunhes-Matuyama polarity change ($t = 0{\cdot}69$ My) by potassium–argon dating (private communication from Dr A. K. Baksi, Queens University, Ontario). All lavas above this horizon are therefore from the Brunhes epoch.

The Canary Islands

The location and names of the major islands in the Canary group are given in Fig. 4. Abdel-Monem, Watkins & Gast (1971, 1972) have presented a summary of the geology of all the major islands and regional crustal genesis, as well as giving an extensive series of potassium–argon age dates on igneous bodies from the major stratigraphic units on all islands. Diagrams of sample sites and surface geology of all islands are included. Table 2 shows all the potassium–argon age results for Brunhes epoch bodies. These radiometric ages provide a reliable framework for identification of Brunhes epoch igneous units on the various islands.

On Lanzarote the Brunhes epoch units were taken from Basaltic Series III and IV of Fuster, Santin & Sagredo (1968a). On Gran Canaria, sampled bodies were from Basaltic Series III of Fuster *et al.* (1968b).

The island of Tenerife is mostly Pliocene or Pleistocene, overlying Lower to

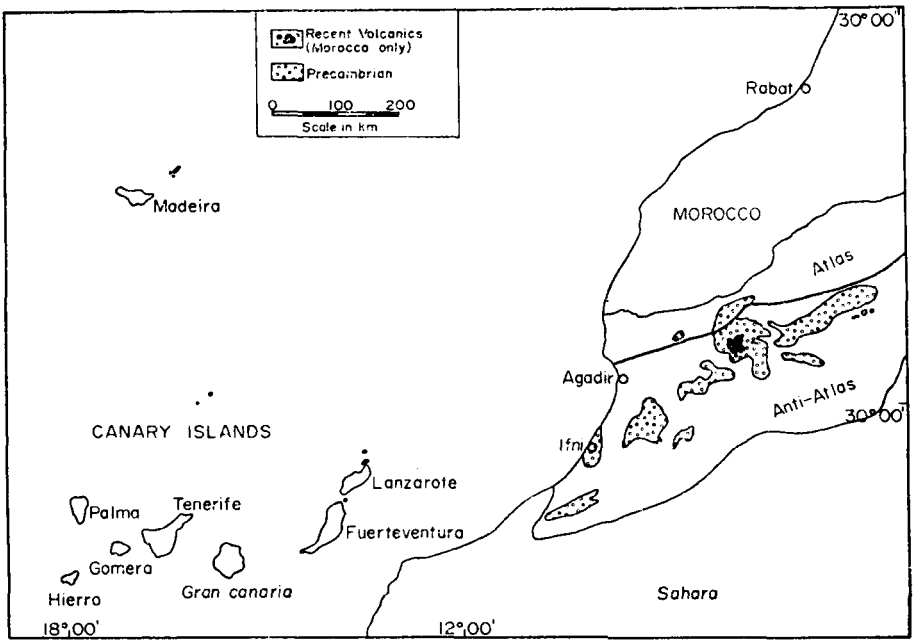


FIG. 4. Map showing locations and names of the Canary Islands.

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Table 2

Potassium–argon age determinations for some of the Brunhes epoch igneous bodies on the Canary Islands

Original Field No.	%K	Ar*/gm $\times 10^{-6}$ Scc.	Ar*/Ar	Age (My)
LZ112	0·74	0·0001	0·006	0·034±0·03
TF104	4·55	0·000	0·000	<0·20
TF114	2·04	0·015	0·05	0·20±0·03
TF 67	2·49	0·052	0·26	0·53±0·06
TF 10	2·81	0·064	0·23	0·56±0·05
TF 88	3·68	0·098	0·62	0·67±0·06
TF112	3·47	0·092	0·58	0·70±0·05
LP15	3·86	0·089	0·04	0·60±0·10
LP27	0·83	0·025	0·15	0·76±0·12
H 4	1·47	0·011	0·04	0·19±0·04
H 7	1·27	0·099	0·12	0·19±0·04
H 9	0·82	0·023	0·18	0·70±0·10
H16	1·53	0·045	0·27	0·74±0·04

For geographic co-ordinates of sample sites, see Table 1.

Ar* = radiogenic argon. Ar = total argon. Decay constants used: $\lambda_\beta = 4{\cdot}72\times10^{-1}\,\mathrm{yr}^{-1}$; $\lambda_e = 0{\cdot}584\times10^{-10}\,\mathrm{yr}^{-1}$, $\lambda_e/\lambda_\beta = 0{\cdot}123$; $K^{40}/\mathrm{K} = 0{\cdot}0119$ atomic per cent. See Abdel-Monem *et al.* (1971, 1972) for other details.

Middle Miocene regions which outcrop in the north-west and eastern peninsulas. The Brunhes epoch samples were taken from the Lower Canadas Series; the Upper Canadas Phonolites; and the Series II Basalts of Fuster *et al.* (1968c).

La Palma consists of a series of basalt flows and minor alkalic intrusives draped around the flanks of a central cone, which has a deep caldera. A diverse igneous succession is exposed in the lower parts of the caldera. The lavas sampled are from the latest Adventive Basaltic Volcanics of Hausen (1969)

All igneous bodies so far sampled on Hierro are of normal polarity Abdel-Monem *et al.* (1972) have shown that the oldest exposures on the island, occurring below a prominent soil horizon just above the base of the Frontera scarp or caldera, may conceivably have been extruded during the Gauss epoch (t = 2·43 to 3·34 My). The younger normal polarity units are all from the Brunhes epoch, and are part of the Tableland Series of Hausen (1964).

Methods

Field methods

The various field methods employed in the previously published results are given in the respective references. These are very variable, and sometimes not specified.

The new data which we shall present here result from field collections made over five years.

None of the collections were made primarily for Brunhes secular variation studies, since in most cases age control was not available prior to the initiation of field studies. Major motives were to study the volcanic stratigraphy, polarity time scale, petrography, geochemistry, and palaeomagnetic pole positions. Results of several of these studies have already been published (Abdel-Monem *et al.* 1971, 1972; Ade-Hall & Watkins 1970; Gunn *et al.* 1970, 1971, 1972; Oversby & Gast 1971; Watkins, Richardson & Mason 1966; Watkins 1972). These independent studies have defined the segment of the collection which is Brunhes in age, to provide the essential framework for the analysis reported in this paper. The number of

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separately oriented specimens taken from each igneous body varies, however, in conflict with idealized requirements of a palaeomagnetic secular variation study.

Doell & Cox (1963) showed that the ultimate accuracy which can be achieved in a palaeomagnetic study of secular variation in a volcanic region is limited by the magnetic anomalies resulting from the terrain over which a given lava flow cools, and the amount of within-flow angular deviation of remanent magnetism directions resulting from experimental (field orientation and laboratory method) limitations. Laboratory measurements, and consideration of the effect of magnetic anomalies of the order of those encountered on Ooshima Island (Rikitake *et al.* 1951), which is a typical volcanic island, yielded estimates of the number of separately oriented cores required to achieve given precision of remanent magnetism directions in volcanic regions. For a within lava precision of K_L [where $K_L = (N-1)/(N-R_L)$, and N = number of separately oriented cores in a lava; R_L = resultant vector, applying unit vector per core] between 100 and 200, the directions can be defined at the 95 per cent confidence level to between 4·75 and 6·25°, if $N = 6$. With $N = 4$, this confidence interval would increase to between 5·5 and 7·5°, whereas for $N = 2$, the interval is 8° to greater than 11°. It is clear therefore that application of palaeomagnetic data to equation (1) would be unlikely to yield S values with very meaningful precision limits if less than four separately oriented examples were employed from each rock unit, and from six to eight samples is clearly desirable. Stone (1970) has argued, on an empirical basis, using results from a palaeomagnetic study of secular variation in Alaskan volcanics, that fewer than six samples may be sufficient to provide acceptable results: he notes that the within-flow scatter does not depend greatly on the number of cores taken. It would seem doubtful, however, if less than four could be employed at a given site to yield results which can be profitably compared with S values from other latitudes. In this communication therefore, we have separated our data into 'first order' results (where $N \geqslant 4$) and 'second order' results (where no lower or upper limits are placed on N).

Cores from the previously unpublished Indian Ocean and Canary Islands studies were all collected using a portable gasoline powered drill and oriented in geographic co-ordinates while still attached to outcrop, to an accuracy of $2\frac{1}{2}°$.

Laboratory

One specimen of 2·3 cm length was taken from each core, which were all of 2·5 cm diameter. All specimens were demagnetized in alternating magnetic fields of $H = 100$, 200 and 300 oersteds (McElhinny 1966), using spinner and astatic magnetometers. Measurement of remanent magnetism direction and intensity was made for the undemagnetized state (the natural remanent magnetism or NRM) and after each demagnetizing treatment.

Computation

The mean direction of magnetization for each lava was computed using the minimum scatter criterion of Irving, Stott & Ward (1961), as discussed by Watkins & Richardson (1968, 1971). This method employs one direction of remanent magnetism from each core following demagnetization treatment, the particular alternating field value used being selected so that the minimum scatter of within-lava directions (or maximum K_L) is produced. This method of computation is preferred to the more commonly applied single demagnetization treatment for a whole sequence of lavas, or for a series of specimens from a single lava, since several studies show that within-lava magnetic stability can vary greatly, so that different demagnetization treatments are required to minimize unstable components within a single lava. Such a computation leads, in practice, to the effective 'weighting' of mean directions in single lavas, to the directions of the more stable specimens, which is clearly desirable in

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practice. This computation method has been criticized by Storetvedt (1970) and Van der Voo (1968, 1971) who prefer to use data resulting from single demagnetization treatments. Van der Voo (1971) calls the minimum dispersion method 'unrealistic', while at the same time stressing that stability differences exist within single lavas: these statements would appear to be somewhat incompatible. In part to counter these criticisms, computations in this presentation are included for single demagnetizing treatments of $H = 100$, 200, and 300 Oe.

Values of S are computed according to equation (1) applying unit vector per lava or point in time, using that direction for each lava which results from the minimum scatter computation. The choice of the reference pole for the computation of δi in equation (1) is problematical: this reference pole can be either the computed mean direction for the given site (to provide S_1); the computed mean directed for the region (to provide S_2), which could be expected to be an improved estimate of the true mean direction since more data and averaging of local anomalies are involved; or the present geographic pole (to provide S_3) since much evidence exists to show that the spin axis was at or very close to its present position during the Brunhes epoch (Doell & Cox 1972). For small values of S, Cox (1969) has shown that, using VGP's,

$$S_T = [2/k]^{\frac{1}{2}} \tag{3}$$

where k = the between-body confidence parameter = $(N-1)/(N-R)$ where N = number of separate bodies, R = the resultant vector applying unit vector per body. In this paper, this method of computation of S_T is designated S_4. Cox (1969b) has provided a method of computing the 95 per cent confidence limits on values of S_4: these are used in this presentation.

Locations for the sites used to derive the previously published data are given in the respective references. The major limitation on the selection of the previously published data involves the nature of the published results: for computation of S (equation (1)) it is necessary to have the VGP or D and I of individual units (or points in time). In some cases, these have been derived from projections showing D and I or VGP. If only mean D, I or VGP for each site is provided, then computation of S cannot be attempted. Age assignments are derived from the geological control provided in the given publications. No data from sedimentary or metamorphic rock units are considered, because of the inherent uncertainty of the accuracy of resulting directions of remanent magnetism.

In Table 3, the mean D, I and VGP co-ordinates are given for the Indian Ocean region results, for each demagnetizing field value, as well as for the minimum dispersion computation method. The mean D and I, and the VGP for each sampled unit resulting from minimum dispersion computation, are included in Table 4.

Table 4(a) shows the results of analysis of those units with $N \geqslant 4$ separately oriented specimens per body (first order data) and Table 4(b) gives the results of the same analyses for all data, regardless of N for each body (second-order data).

Table 5(a) presents the results of computation of the angular dispersion (S) for each site or group of sites, for the first order data, using the various reference poles (S_1, S_2, S_3 and S_4) described above, as well as the upper and lower limits of the associated 95 per cent confidence intervals (Su and Si). Table 5(b) presents the corresponding results for second order data.

Discussion

Minimum dispersion computation

Inspection of Table 3 reveals two facts about the minimum dispersion computation method, as applied to the data for the Indian Ocean studies: the mean D and I values following each demagnetizing treatment are subject to very small changes;

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Table 3

Palaeomagnetic results for Indian Ocean Islands for different demagnetization treatments and minimum dispersion computation

Site	H	$D°$	$I°$	R	K	$\alpha(95°)$	θ'	ϕ'	$dp°$	$dm°$
Anjouan	100	2·3	−24·5	23·7	18·3	6·9	87·6	150·2	4·0	7·5
	200	4·3	−21·7	24·0	25·1	5·9	85·7	121·7	3·3	6·2
	300	4·1	−21·3	24·1	25·4	5·9	85·8	118·6	3·3	6·2
	max	3·9	−21·8	24·2	26·9	5·7	86·1	121·7	3·2	6·0
Grande Comore	100	358·8	−22·6	13·7	41·6	5·8	88·8	299·6	3·3	6·2
	200	1·0	−22·7	12·7	40·1	6·1	89·0	146·2	3·5	6·5
	300	359·1	−21·1	13·7	43·8	5·7	89·0	348·4	3·1	5·9
	max	359·8	−21·6	13·7	42·2	5·8	89·6	20·7	3·2	6·1
Amsterdam	100	6·1	−54·4	11·4	18·7	9·3	84·4	140·1	9·3	13·2
	200	9·2	−54·7	11·4	17·9	9·5	82·2	151·4	9·5	13·3
	300	12·0	−54·2	11·3	16·3	10·0	79·9	154·3	9·9	14·1
	max	9·9	−54·5	11·3	17·0	9·8	81·6	152·2	9·8	13·8
Possession	100	0·0	−63·1	36·1	42·1	3·5	88·1	51·3	4·4	5·6
	200	359·1	−63·4	36·0	37·3	3·9	88·3	30·7	4·9	6·2
	300	356·7	−63·1	35·9	34·2	4·1	87·1	0·3	5·1	6·4
	max	4·0	−62·0	36·2	20·8	5·2	85·7	94·6	6·3	8·1
East Island	100	4·8	−61·5	41·7	31·6	3·8	84·8	95·1	4·5	5·9
	200	7·2	−60·0	41·3	25·1	4·3	82·3	97·7	4·3	4·1
	300	12·1	−62·0	35·0	17·8	5·5	80·8	125·6	6·6	8·5
	max	7·5	−61·7	41·6	31·2	3·8	83·5	110·6	4·6	5·9

H = demagnetizing field treatment (max = minimum dispersion computation: see text). For explanation of symbols see caption to Table 1. Also: $K = N-1/N-R$; where R = resultant vector, applying unit vector per body. dp and dm = length of semi axes of 95 per cent confidence oval, in line with and normal to great circle between site and mean pole position.

and the between-lava precision parameter (K) is *not* always at a maximum following minimum dispersion computation, but rather following a single demagnetizing treatment. (In the Possession Island data, for example (Table 3), K is maximum after demagnetization of all specimens in a field of 100 Oe, and is less for the minimum dispersion computation than at any of the three demagnetizing treatments. The data for Anjouan is quite different, with K being maximum at the minimum dispersion computation.). This is because the mean D and I values *within* each lava following minimum dispersion computation may differ from those resulting from single demagnetizing treatments, so that when *between* lava means are computed, the minimum dispersion data can yield wider confidence parameters. Accurate values of K are especially important in secular variation studies. Selection of results for a site or group corresponding to a maximum *between-lava* K has no physical basis: there is no reason whatsoever to believe that the true mean direction of remanent magnetism in a suite of lavas is that corresponding to maximum K values: there would clearly exist a strong possibility that a misleadingly low S value could result from application of this data computation method. This is to be contrasted with the maximum K_L computation system, which (as argued above) is based on the reality of stability variations within single lavas. This is not to say that minimum dispersion computation is incapable of yielding misleading data for a single body: Watkins & Richardson (1971) have argued that this will not occur, however, if at least one specimen in the given body is of medium to high stability to 'weight' the mean direction. Also the effect of imperfect demagnetizing procedures (whereby spurious components are added to the specimen) is eliminated by minimum dispersion computation provided cores within a single lava are drilled at varying azimuths, since any systematic component addition to each core will then result in a decreasing K_L value

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Table 4

Mean directions, pole positions, and precision parameters for all Brunhes epoch palaeomagnetic data in the Indian and Atlantic regions, by site, group, and region

(a) First order data ($n \geq 4$ samples per body)

No.	Site or group	N	B	D	I	Analysis of directions K_1	ϕ^1	θ^1	$\alpha(95)_1$	K_2	Analysis of VGP ϕ^{11}	θ^{11}	$\alpha(95)_2$
1	Grande Comore	14	84	359·8	−21·6	43·5	25·8	89·5	6·2	55·0	19·5	89·8	5·0
2	Anjouan	23	137	4·2	−22·4	25·2	127·2	85·8	6·1	32·8	130·6	86·2	5·1
1+2	Comores Is.	37	221	2·5	−22·1	29·9	128·7	87·5	4·4	38·8	129·1	87·7	3·7
3	Amsterdam	16	72	9·3	−53·4	16·1	143·1	81·5	11·7	12·0	155·2	81·9	12·2
4	East Island	43	156	7·5	−61·7	31·2	110·3	83·6	3·9	16·9	125·1	84·7	5·2
5	Possession	36	176	4·8	−62·3	20·4	103·2	85·5	5·4	16·9	119·7	86·9	5·7
4+5	Crozet Is.	79	332	6·3	−62·0	25·4	107·4	84·5	3·2	17·1	123·3	85·7	3·8
1 to 5	Indian Ocean	127	625	—	—	—	—	—	—	19·7	129·5	86·2	2·8
6	France	24	237	356·2	+62·4	34·5	238·8	86·7	5·1	19·0	266·8	87·0	6·6
7	Spain	6	70	352·1	+54·2	42·3	224·8	80·3	10·4	25·7	227·2	80·5	11·3
8	Turkey	13	110	6·8	+56·8	50·5	147·6	84·4	6·2	35·0	133·6	85·2	6·8
9	Tenerife	14	75	357·8	+31·7	7·4	174·4	78·6	16·4	14·0	178·1	80·0	10·4
10	La Palma	15	80	2·3	+39·9	23·8	143·0	83·6	8·0	23·7	143·4	84·3	7·4
11	Hierro	10	50	4·7	+47·2	18·7	62·0	85·8	11·5	20·3	43·8	86·1	9·8
9 to 11	Canary Is.	40	205	2·0	+39·3	13·2	146·7	83·6	6·5	18·9	142·4	84·6	4·9
6 to 11	Atlantic	83	622	—	—	—	—	—	—	20·2	170·9	86·8	3·4
1 to 11	Indian & Atlantic	210	1247	—	—	—	—	—	—	20·2	146·1	86·7	2·1

(b) Second order data (any number of samples per body)

1	Ethiopia	6	6	5·4	+14·5	39·3	239·3	85·2	10·8	85·9	240·0	85·3	6·2	Pouchan & Roche; 1971
2	Grande Comore	14	84	359·8	−21·6	42·5	25·8	89·5	6·2	55·0	19·5	89·8	5·0	Hajash 1970
3	Anjouan	25	142	3·9	−21·8	26·9	121·7	86·1	5·7	35·2	124·8	86·4	4·7	,,
2+3	Comores Is.	39	226	2·4	−21·8	31·2	113·6	87·5	4·2	40·6	123·2	87·7	3·5	,,
4	Mauritius	8	22	359·5	−45·5	77·2	241·6	83·4	6·3	72·4	243·6	82·8	5·8	McDougall & Chamalaun 1969
5	Reunion	49	59	6·3	−34·2	16·6	156·2	84·1	5·1	17·0	183·9	83·2	4·8	Chamalaun 1968
6	Amsterdam	12	75	9·9	−54·5	17·0	151·6	81·6	10·8	12·7	164·5	81·7	11·3	Gunn *et al.* 1971

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Table 4 (*continued*)

No.	Site or group	N	B	D	I	Analysis of directions K_1	ϕ^1	θ^1	$\alpha(95)_1$	Analysis of VGP K_2	ϕ^{11}	θ^{11}	$\alpha(95)_2$	References
7	East Island	43	156	7·5	−61·7	31·2	110·3	83·6	3·9	16·9	125·1	84·7	5·2	Gunn *et al.* 1970, 1971
8	Possession	38	181	4·0	−62·0	20·8	94·6	85·7	5·2	16·9	108·4	87·1	5·5	Gunn *et al.* 1972
7+8	Crozet Is.	81	337	5·9	−61·8	25·5	104·7	84·6	3·2	17·1	119·6	85·9	3·8	
9	Heard	8	19	354·7	−61·8	39·1	52·2	79·3	9·0	19·9	48·9	81·7	11·1	Irving *et al.* 1965; Stephenson 1972
1 to 9	Indian Ocean	203	744	—	—	—	—	—	—	19·6	152·0	86·5	2·2	
10	Jan Mayen	10	10	339·3	+82·6	52·2	306·9	82·7	6·7	16·2	335·1	81·7	11·0	Fitch *et al.* 1965
11	Germany	6	11	27·3	+68·8	53·1	80·8	73·0	9·3	21·0	79·2	73·7	12·5	Nairn 1962
12	France	24	237	356·2	+62·4	34·5	238·8	86·7	5·1	19·0	266·8	87·0	6·6	Doell 1970
13	Spain	7	73	356·2	+53·5	37·9	203·9	81·3	9·9	23·0	207·1	81·6	11·0	Guardia 1964
14	Azores	22	22	3·2	+58·2	30·4	63·8	87·8	5·1	26·9	76·5	88·6	5·8	Serughetti & Roche 1968
15	Turkey	17	121	5·4	+55·6	45·8	165·5	84·8	5·3	34·8	146·2	85·7	5·7	Sanver 1968
16	Lanzarote	21	50	17·6	+37·7	38·1	99·0	72·2	5·2	35·6	98·5	72·5	5·1	Abdel-Monem *et al.* 1971, 1972
17	La Palma	16	83	1·3	+40·8	23·4	150·9	84·5	7·8	23·0	153·5	85·3	7·3	,,
18	Tenerife	21	84	359·1	+30·4	10·6	168·0	77·9	10·2	18·5	170·2	78·2	7·1	,,
19	Gran Canaria	5	10	24·9	+19·0	6·6	51·8	63·4	32·6	22·4	108·0	61·9	22·4	,,
20	Hierro	34	99	8·0	+48·0	16·4	59·6	82·8	6·3	15·6	46·3	82·7	6·1	,,
16 to 20	Canary Is.	97	326	9·6	+39·3	13·5	111·2	80·7	4·0	15·5	107·7	81·9	3·6	
21	Argentina	6	12	357·6	−60·5	52·9	131·5	85·1	9·3	24·2	130·5	84·9	11·6	Creer & Valencio 1969
22	Tristan	9	27	359·8	−50·1	53·8	345·6	83·8	7·1	41·4	348·7	84·2	7·2	Creer 1964
10 to 22	Atlantic	198	839	—	—	—	—	—	—	17·5	104·5	85·7	2·4	

Number refers to data group. Site: see text. N = number of separate bodies sampled. B = total number of samples from N bodies. D and I = mean declination and inclination of remanent magnetism for given site applying unit vector per body (Table 1); D is in degrees east of north, and I is in degrees with respect to horizontal (+ = below horizontal, − = above horizontal); $K_1 = (N-1)/(N-R)$ where R_1 = resultant vector, applying unit vector per body to directions; ϕ', θ' = longitude and latitude of mean virtual geomagnetic pole, calculated using directions for each body. $K_2 = (N-1)/(N-R_2)$ where R_2 = total resultant vector, applying unit vector per body to virtual geomagnetic poles (VGP). ϕ'' and θ'' = longitude and latitude of mean VGP, calculated using VGP for each body. $\alpha(95)_1$ and $\alpha(95)_2$ = semi vertical angle of 95 per cent confidence oval, calculated using unit vector for each direction, and unit vector for each VGP, respectively. Note: (a) Mauritius results are for eight bodies only, although the result given by McDougall & Chamalaun (1969) is for 13 units (five of which are not tabulated in detail) (b) Data for France excludes results for seven additional sampled bodies, since no demagnetizing treatments were associated with the results given (Doell 1970). (c) Results from Ethiopia and the Azores derived from D and I for each body, and VGP for each body, respectively, as read from published projections of the results.

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Table 5

Longitude and latitude of site, total angular standard deviations for various reference points, and confidence limits, for testing of geomagnetic field models (Fig. 1)

(a) First order data ($n \geqslant 4$ samples per body)

Number	Site or group	$\phi°$	$\theta°$	S_1	S_2	S_3	S_4	Su	Si
1	Grande Comore	43·3	−11·7	10·9	11·0	11·6	10·9	14·7	8·6
2	Anjouan	44·4	−12·2	14·8	14·2	14·2	14·2	17·8	11·8
1+2	Comores Is	43·7	−11·9	13·3	13·1	13·2	13·0	15·4	11·2
3	Amsterdam	77·6	−37·8	25·4	23·8	24·5	24·4	35·3	18·8
4	East Island	52·2	−46·5	20·6	19·9	19·9	19–7	23·1	17·3
5	Possession	51·7	−46·5	20·5	20·2	20·2	19·7	23·5	16·9
4+5	Crozet Is.	51·9	−46·5	20·4	19·9	19·9	19·6	22·1	17·6
6	France	3·0	+45·7	19·1	18·8	19·4	18·6	23·3	15·5
7	Spain	2·5	+42·2	19·1	16·0	18·3	16·0	26·3	11·5
8	Turkey	43·0	+39·0	14·6	13·8	13·8	13·7	18·8	10·8
9	Tenerife	343·8	+28·4	24·5	22·1	23·3	22·3	30·1	17·7
10	La Palma	342·3	+28·7	17·7	16·8	17·1	16·7	22·3	12·0
11	Hierro	342·0	+27·7	18·6	18·1	19·3	18·0	25·9	13·8
9 to 11	Canary Is.	343·8	+28·4	19·6	19·3	19·4	18·6	22·1	16·1

(b) Second order data: any number of samples per body

Number	Site or group	$\phi°$	$\theta°$	S_1	S_2	S_3	S_4	Su	Si
1	Ethiopia	42·5	+12·0	10·1	8·8	10·8	8·7	14·4	6·2
2	Grande Comore	43·3	−11·7	11·0	11·8	11·6	11·0	14·8	8·7
3	Anjouan	44·4	−12·2	14·2	13·7	13·8	13·7	17·0	11·4
2+3	Comores Is.	43·7	−11·9	13·0	12·8	12·9	12·8	15·2	11·1
4	Mauritius	57·5	−20·3	12·2	9·5	12·9	9·5	14·0	7·1
5	Reunion	55·5	−21·2	21·1	19·9	20·7	19·6	22·8	17·2
6	Amsterdam	77·6	−37·8	24·8	23·1	23·8	23·4	32·6	18·4
7	East Island	52·2	−46·5	20·6	19·9	20·1	19·7	23·1	17·3
8	Possession	51·7	−46·5	20·4	20·1	20·3	19·7	23·4	17·0
7+8	Crozet Is.	51·9	−46·5	20·4	19·9	20·0	19·6	22·0	17·7
9	Heard	73·5	−53·1	20·3	18·3	20·9	18·3	27·8	13·7
10	Jan Mayen	351·8	+71·1	22·1	20·3	23·6	20·2	29·1	15·4
11	Germany	7·0	+50·5	25·2	25·2	22·5	17·7	29·2	12·7
12	France	3·0	+45·7	19·1	18·8	20·2	18·6	23·3	15·5
13	Spain	2·5	+42·2	25·2	16·7	20·2	17·0	26·7	12·2
14	Azores	328·8	+39·6	15·8	15·7	16·0	15·7	19·8	13·0
15	Turkey	43·0	+39·0	14·5	13·8	14·1	13·8	18·3	11·1
16	Lanzarote	346·5	+29·1	22·4	13·7	19·1	13·6	17·3	11·2
17	La Palma	342·3	+28·7	17·7	17·0	17·4	16·9	22·5	13·6
18	Tenerife	343·8	+28·4	22·4	18·9	21·7	18·8	23·9	15·6
19	Gran Canaria	344·4	+28·0	42·8	29·3	39·5	17·2	30·2	12·0
20	Hierro	342·0	+27·7	22·0	20·7	21·6	20·5	24·6	17·5
16 to 20	Canary Is.	343·8P	+38·4	22·6	20·8	21·2	20·6	23·0	18·7
21	Argentina	290·0	−37·0	17·5	16·6	16·8	16·5	27·2	11·8
22	Tristan	347·3	−37·1	14·0	12·6	15·5	12·6	18·6	9·5

Number refers to data group. Site: see text. ϕ and θ = mean longitude and latitude of site. Longitude is in degrees east. Latitude is negative for southern latitudes; and positive for northern latitudes. $S = (N-1)^{-\frac{1}{2}} \left(\sum_{i=1}^{N} \delta i^2 \right)^{\frac{1}{2}}$ where δi = angular difference between VGP for the body and the reference pole. Suffixes to S define reference pole: 1 = δi with respect to geographic pole; 2 = δi with respect to mean pole, for site or group calculated using VGP's (Table 2); 3 = δi with respect to mean pole position for Atlantic and Indian Oceans calculated using VGP's; $S_4 = (2/k)^{\frac{1}{2}}$; Su and Si = upper and lower confidence parameters for S_4 (Cox 1969b).

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for a lava. We find it impossible to accept the ' unrealistic ' description of the minimum dispersion computation method (Van der Voo 1971); and we therefore choose to analyse basic data by the minimum dispersion method.

S: Second order data

Fig. 5 shows the values of S_4 and associated 95 per cent confidence limits as a function of geographic latitude for all second order data. The corresponding latitude dependence of S_D (or D), S_N, A and S_T (or C) as functions of latitude, as discussed earlier (Fig. 1), are superimposed on the results.

Inspection of Fig. 5 shows that the confidence limits on most of the S_4 values are very large. This almost certainly reflects the low number of separate samples from each lava (yielding low K_L) values), and/or low numbers of separate bodies, or points in time (Table 4(b)). We believe that the result from Reunion (Fig. 5) is

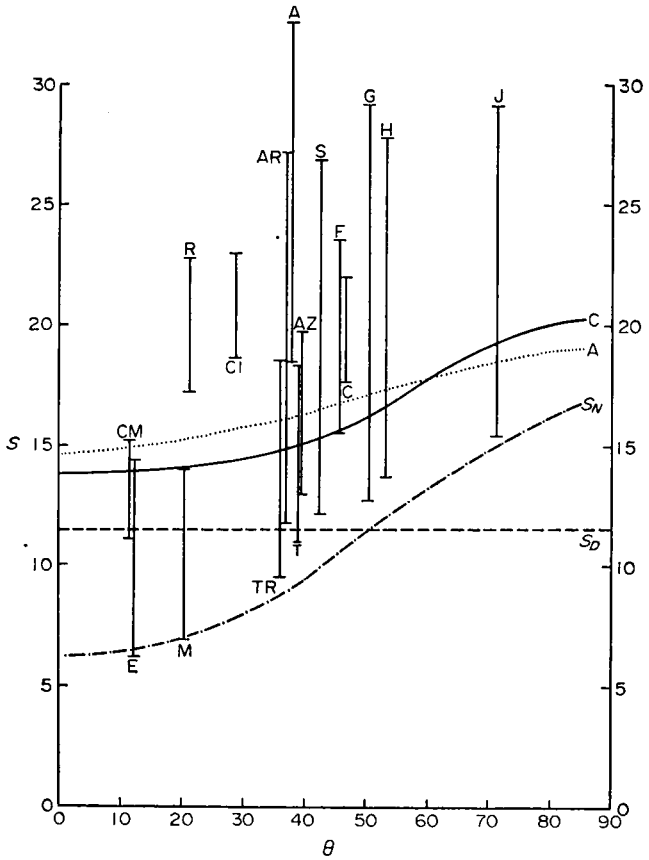


FIG. 5. Relation between geographic latitude (θ) and angular dispersion (S) of Brunhes epoch virtual geomagnetic poles for all second order data (Tables 4 and 5: any number of separately oriented samples per body). Results given in form of 95 per cent confidence limits (Cox 1969b). S_D, S_N, A and C, correspond to $S : \theta$ relationships for various geomagnetic models and present field behaviour (see text and caption to Fig. 1 for definitions). Results are for (in order of increasing latitude): CM = Comores; E = Ethiopia; M = Mauritius; R = Reunion; CI = Canary Islands; TR = Tristan; AR = Argentina; A = Amsterdam Island; T = Turkey; AZ = Azores; S = Spain; F = France; C = Crozet; G = Germany; H = Heard Island; J = Jan Mayen.

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instructive in this context: while a large number of separate bodies (49) are involved to yield smaller 95 per cent confidence parameters, only one sample per lava comprise most of the data, resulting in a high S_4 value. Of course, any result can be suspected to be a function if insufficient time range (if S_4 is small) or insufficient separate bodies (if S_4 is large), illustrating some of the dilemmas of palaeomagnetic secular variation studies: it is clearly essential to obtain wide time coverage. It is somewhat surprising that despite the unsatisfactory nature of this second order data, a latitude dependence of S_4 is nevertheless apparent from inspection of all data in Fig. 5. This relationship, while poorly defined, is sufficient to remove the possibility of the dipole wobble model (S_D in Fig. 5) being the cause of the Brunhes epoch geomagnetic field.

S: First order data

Fig. 6 illustrates the 95 per cent confidence intervals of S_4 for the first order results (Tables 4(a) and 5(a)) for those results involving at least 15 separate bodies. This selectivity should provide the maximum possibility of avoiding the problems in time range and within-lava K_L values, discussed above. A clear latitude dependence of S_4

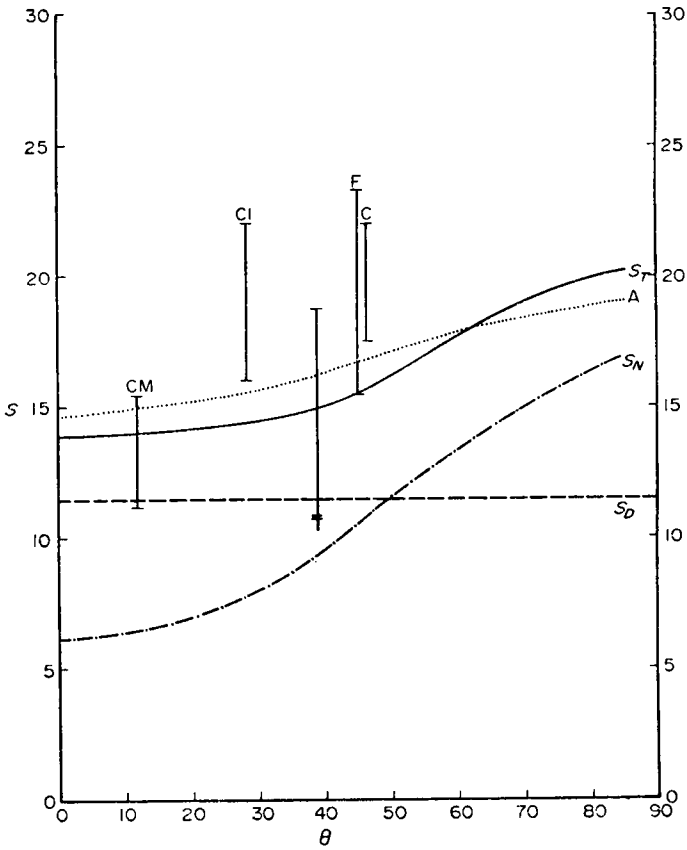


FIG. 6. Relation between geographic latitude (θ) and angular dispersion (S) of Indian and Atlantic ocean (Brunhes Epoch) virtual geomagnetic poles, for all first order data (Tables 4 and 5, four or more separately oriented samples per body), when 15 or more separate bodies are involved. Results are given in form of 95 per cent confidence limits (Cox 1969b). For explanation and definitions of S_D, S_N, A, and S_T see text and caption to Fig. 1. Results are for (in order of increasing latitude): CM = Comores; CI = Canary Islands; T = Turkey; F = France; C = Crozet.

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is evident, and there appears to exist the possibility that S increased more rapidly with latitude during the Brunhes in the Atlantic and Indian Ocean regions, than at present (Fig. 1). Since Doell (1969) has shown that the best estimate of the Brunhes dipole wobble is about $11\frac{1}{2}°$, then it follows that non-dipole activity would have to be more intense at the higher latitudes of the Atlantic and Indian Ocean Regions, to explain the results in Fig. 6. As with the second order data, the dipole wobble model is eliminated from further consideration. Models A and C cannot be distinguished by the data presented here, and it would appear doubtful that those models can be distinguished palaeomagnetically, unless ideal sequences of materials can be sampled at very high latitudes. This suggests that since it appears very probable that the only palaeomagnetically distinguishable geomagnetic field model is S_D (or dipole wobble), and since this is clearly rejected by both selected first and second order results, then in this context nothing was gained by taking four or more cores per body, as opposed to two per body. Our results show, however, that detection of different latitudinal changes of S, or non-dipole minima (Fig. 1) cannot be made with low K_L values, in agreement with Doell & Cox (1963).

The data from the Comores (Fig. 6) in particular, and also the second order results from Ethiopia and Mauritius (Fig. 5) pose the intriguing possibility that Brunhes subdued non-dipole activity may be a feature of the West central Indian Ocean, in similar fashion to the central Pacific Ocean feature (Fig. 2) defined by Doell & Cox (1971). At this time, no distinction can be made between an attenuated non-dipole field superimposed on dipole wobble, or models A and C, for the Brunhes epoch in the West central Indian Ocean.

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References

Abdel-Monem, A., Watkins, N. D. & Gast, P. W., 1971. Potassium-argon ages, volcanic stratigraphy, and geomagnetic polarity history of the Canary Islands: Lanzarote, Fuerteventura, Gran Canaria, and La Comera, *A. J. Sci.*, **271,** 490–521.

Abdel-Monem, A., Watkins, N. D. & Gast, P. W., 1972. Potassium–argon ages, volcanic stratigraphy, and geomagnetic polarity history of the Canary Islands: Tenerife, La Palma, and Hierro, *A. J. Sci.*, in press.

Ade-Hall, J. M. & Watkins, N. D., 1970. Absence of correlations between opaque petrology and natural remanence polarity in Canary Island lavas, *Geophys. J. R. astr. Soc.*, **19,** 351–360.

Cox, A., 1962. Analysis of present geomagnetic field for comparison with paleomagnetic results, *J. Geomagn. Geoelect., Kyoto.* **13,** 101–122.

Cox, A. 1969a. A paleomagnetic study of secular variation in New Zealand, *Earth. Planet. Sci. Lett.*, **6,** 257–267.

Cox, A., 1969b. Confidence limits for the precision parameter *K*, *Geophys. J. R. Astr. Soc.*, **18,** 545–549.

Chamalaun, F. H., 1968. Paleomagnetism of Reunion Island and its bearing on secular variation, *J. geophys. Res.*, **73,** 4647–4659.

Creer, K. M., 1964. Paleomagnetic measurements on lavas from Tristan and Inaccessible Island, *Phil. Trans. R. Soc. Lond A*, **256,** 569–573.

Creer, K. M., Irving, E. & Nairn, A. E. M., 1959. Palaeomagnetism of the Great Whin Sill, *Geophys. J. R. astr. Soc.*, **2,** 306–323.

Creer, K. M. & Valencio, D. A., 1969. Palaeomagnetic and rock magnetic studies on the Cenozoic basalts from Western Argentina, *Geophys. J. R. astr. Soc.*, **19,** 113–146.

Doell, R. R., 1969. Paleomagnetism of the Kauai Volcanic Series, Hawaii, *J. geophys. Res.*, **74,** 4857–4866.

Doell, R. R., 1970. Paleomagnetic secular variation study of lavas from the Massif Central, France, *Earth Planet. Sci. Lett.*, **8,** 352–362.

Doell, R. R. & Cox, A., 1963. The accuracy of the paleomagnetic method as evaluated from Historic Hawaiian lava flows, *J. geophys. Res.*, **68,** 1997–2009.

Doell, R. R. & Cox, A., 1971. Pacific geomagnetic secular variation, *Science*, **171,** 3377–3405.

Doell, R. R. & Cox, A., 1972. Pacific geomagnetic secular variation, in *The Nature of the Solid Earth* (ed. E. C. Robertson) McGraw-Hill (New York), in press.

Esson, J., Flower, M. G. J., Strong, D. F., Upton, B. G. J. and Wadsworth, W. J., 1970. Geology of the Comores Archipelago, Western Indian Ocean, *Geol. Mag.*, **107,** 549–557.

Fisher, R., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.*, **A217,** 295–305.

Fitch, F. J., Nairn, A. E. M. & Talbot, C. J., 1965. Palaeomagnetic studies on rocks from North Jan Mayen, *Norsk Polar institutt Årbok* (1963), 49–60.

Fuster, J. M., Santin, S. F. & Sagredo, A., 1968a. Geology and volcanology of the Canary Islands, Lanzarote, *Inst. 'Lucas Mallada', Madrid, Spec. Publ. for Int. Symp. Volc,. Tenerife,* 177 pp.

Fuster, J. M., Hernandez-Pacheco, A., Muroz, M., Badiala, E. R. & Cacho, L. G., 1968b. Geology and volcanology of the Canary Islands, Gran Canaria, *Inst. 'Lucas Mallada', Madrid, Spec. Publ. for Int. Symp. Volc., Tenerife,* 243 pp.

Fuster, J. M., Arana, V., Brandle, J. L., Navarro, M., Alonso, U. & Appaiicio, A., 1968c. Geology and volcanology of the Canary Islands, Tenerife, *Inst. 'Lucas Mallada', Madrid, Spec. Publ. for Int. Symp. Volc., Tenerife,* 218 pp.

Guardia, P., 1964. Volcans Tertiares et Quaternaires de la Province de Gerone et paleomagnétisme de leurs coulées, *Rev. Geog. Phys. Geol. Dynam.*, **6,** 201–210.

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Gunn, B. M., Coy-Yll, R., Watkins, N. D., Abranson, C. E. & Nougier, J., 1970. Geochemistry of an oceanite–ankaramite–basalt suite from East Island, Crozet Archipelago, *Contrib. Min. Pet.*, **28,** 319–339.

Gunn, B. M., Nougier, J., Abranson, C. E., Watkins, N. D. & Hajash, A., 1971. New Amsterdam, an isolated volcano in the southern Indian Ocean, *Contrib. Min. Pet.* **32,** 79–92.

Gunn, B. M., Abranson, C. E., Watkins, N. D. & Nougier, J., 1972. Geochemistry of the Crozet archipelago: a summary. *Proc. Second Conf. on Antarctic Geology and Geophysics*, ed. (Oslo, 1970) R. J. Adie, in press.

Hajash, A., 1970. *A Paleomagnetic investigation of the Comores Islands, West-central Indian Ocean, Unpublished M. S. Thesis*, Florida State University.

Hausen, H., 1964. Rasgos geologicos genorales de la Isla del Hierro; Patronato de la " Casa de Colon ", *Annuario de Estudios Atlanticos (Madrid)*, **10,** 45 pp.

Hausen, H., 1969. Some contributions to the geology of La Palma, *Comment Soc. Sci. Phys. Math. Fenn.*, **35,** 1–140.

Irving, E. & Ward, M. A., 1964. A statistical model of the geomagnetic field *Geofis. Pura Appl.* **57,** 25–30.

Irving, E., Stott, P. M. & Ward, M. A., 1961. Demagnetization of igneous rocks by alternating magnetic fields, *Phil. Mag.*, **6,** 225–241.

Irving, E., Stephenson, P. J. & Major A., 1965. Magnetism in Heard Island rocks, *J. geophys. Res.*, **70,** 3421–3427.

McDougall, I. & Chamalaun, F. H., 1969. Isotopic dating and geomagnetic polarity studies on volcanic rocks from Mauritius, Indian Ocean, *Bull. geol. Soc. Am.*, **80,** 1419–1442.

McElhinny, M. W. 1966. An improved method of demagnetizing rocks in alternating magnetic fields, *Geophys. J. R. astr. Soc.*, **10,** 369–374.

Nairn, A. E. M., 1962. Paleomagnetic investigation of the Tertiary and Quaternary igneous rocks I: Preliminary collections in the Eifel, Siebengebirge, and Westerwald, *Not. Hessischen Landesmtes für Bodenforschung zu Wiesbaden*, **90,** 412–424.

Oversby, V. & Gast, P. W., 1971. Isotopic composition of lead in volcanic rocks from Tenerife, Canary Islands, *J. geophys. Res.*, **76,** 3402–3413.

Pouchan, P. & Roche, A., 1971. Etude paleomagnetique de formation volcaniques due Territoire de Afars et Issas, *C. r. hebd. Seanc. Acad. Sci. Paris*, **272,** 531–534.

Rikitake, T., Yokoyama, A., Okada, A. & Hishiyama, Y., 1951. Geomagnetic studies of Volcano Mihara. The third paper. (Magnetic survey and continuous observation of changes in geomagnetic declination), *Bull. Earthquake Res. Inst., Tokyo Univ.*, **29,** 584–604.

Saint Ours, J. de., 1960. *Etudes geologiques dans l'extreme nord de Madagascar et L'Archipel des Comores*, Service Geologique, Tananarive.

Sanver, M., 1968. A palaeomagnetic study of Quaternary volcanic rocks from Turkey, *Phys. Earth Planet. Int.* **1,** 403–421.

Serughetti, J. & Roche, A., 1968. Etude paleomagnétique de laves de l'ile de Flores, *C. r. hebd, Seanc. Acad. Sci. Paris*, **B 267,** 1185–1188.

Stephenson, P. J., 1972. The geochemistry of some Heard Island igneous rocks, *Proc. Second Conf. on Antarctic Geology and Geophysics*, R. J. Adie, in press.

Stone, D., 1970. Paleosecular variation in Alaska, pp 117–126 in *Paleogeophysics*, ed. S. K. Runcorn, Academic Press, New York.

Storetvedt, K. M., 1970. Palaeomagnetism of the Lisbon Volcanics: a discussion of a recent paper by N. D. Watkins and A. Richardson, *Geophys J. R. astr. Soc.* **19,** 107–110.

Strong, D. F. & Flowers M. E. J., 1969. The significance of sandstone inclusions in lavas of the Comores Archipelago, *Earth Planet. Sci Lett.*, **7,** 47–50.

Van der Voo, R., 1968. Comments on ' Palaeomagnetism of the Lisbon Volcanics' by N. D. Watkins and A. Richardson, *Geophys. J. R. astr. Soc.*, **16,** 543–548

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Van der Voo, R., 1971. Renewed palaeomagnetic study of the Lisbon volcanics and implications for the rotation of the Iberian peninsula, *J. geophys. Res.*, **76,** 3913–3921.

Watkins, N. D.& Richardson, A., 1968l Palaeomagnetism of the Lisbon Volcanics, *Geophys. J R. astr. Soc.*, **15,** 287–304.

Watkins, N. D. & Richardson, A., 1971. Reply to K. M. Storetvedt's ' Comments on ' Palaeomagnetism of the Lisbon Volcanics ', '*Geophys. J. R. astr. Soc.*, **22,** 446–448.

Watkins, N. D., 1972. Hemispherical contrasts in support for the offset dipole hypothesis during the Brunhes Epoch: the case for an unequal co-axial dipole pair as a possible geomagnetic field source, *Geophys. J. R. astr. Soc.*, **28,** 193.

Watkins, N. D., Richardson, A. & Mason R. G., 1966. Paleomagnetism of the Macaronesian insular region: the Canary Islands, *Earth Planet. Sci. Lett.*, **1,** 225–231.

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